

The original version had been years in the making, and expectations for the next release were incredibly high. But the longer we worked, the more afraid we became of how customers would react when they finally saw the new version. As our plans became more ambitious, so too did the number of bugs, conflicts, and problems we had to deal with. Pretty soon we got into a situation in which we could not ship anything. Our launch date seemed to recede into the distance. The more work we got done, the more work we had to do. The lack of ability to ship eventually precipitated a crisis and a change of management, all because of the trap of large batches.

These misconceptions about batch size are incredibly common. Hospital pharmacies often deliver big batches of medications to patient floors once a day because it's efficient (a single trip, right?). But many of those meds get sent back to the pharmacy when a patient's orders have changed or the patient is moved or discharged, causing the pharmacy staff to do lots of rework and reprocessing (or trashing) of meds. Delivering smaller batches every four hours reduces the total workload for the pharmacy and ensures that the right meds are at the right place when needed.

Hospital lab blood collections often are done in hourly batches; phlebotomists collect blood for an hour from multiple patients and then send or take all the samples to the lab. This adds to turnaround time for test results and can harm test quality. It has become common for hospitals to bring small batches (two patients) or a single-patient flow of specimens to the lab even if they have to hire an extra phlebotomist or two to do so, because the total system cost is lower.⁷

PULL, DON'T PUSH

Let's say you are out for a drive, pondering the merits of small batches, and find yourself accidentally putting a dent in your new 2011 blue Toyota Camry. You take it into the dealership for repair and wait to hear the bad news. The repair technician tells you that